Reading the Body: A Calibrated, Abstaining Classifier for 13-Class News Topic Assignment

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***Abstract*—News topic classifiers are conventionally trained on headlines, because that is the text a feed provides. This work tests whether the full article body is worth the cost of scraping it, on a purpose-built corpus of 14,189 articles from 40 publishers with 8,001 human labels across 13 topics. It is: reading the first 4,000 characters of the body instead of the headline and summary is worth $+0.059$ macro-F1 ($0.712 \to 0.771$), with non-overlapping bootstrap intervals and McNemar $p = 7.5 \times 10^{-6}$. Every alternative modelling choice measured against that linear baseline — entity scrubbing, gradient-boosted trees, random forests, sentence embeddings and four ensembles — lost or tied, so the accuracy came from data preparation rather than from the classifier. The remaining contribution is a system that knows when it is unsure: isotonic calibration turns an uncalibrated SVM margin into a probability with expected calibration error 0.021, and a single confidence threshold lets the model decline the hardest 19% of articles, raising accuracy on what it does file from 77.7% to 83.4%. On a test split opened exactly once, macro-F1 is 0.751 $[0.719, 0.780]$.**

***Index Terms*—text classification, probability calibration, classification with rejection, support vector machines, TF-IDF, news**

I. Introduction

A reader following several newspapers sees the same story repeatedly, filed under whatever section each publisher happened to choose. Automatic topic assignment is the prerequisite for reorganising that stream. The standard approach classifies from the headline, because an RSS item usually carries nothing else.

That constraint is now soft: a collector that follows the article link has the whole body, roughly $100\times$ more text. Whether the extra text pays is an empirical question rather than an obvious one — bodies carry publisher house style, boilerplate and author biographies, all of which are shortcuts a classifier can learn instead of the topic.

This paper reports three results.

1) The body is worth $+0.059$ macro-F1, significant on every linear model tested (Section V).
2) Seven alternative model families and four ensembles were measured against a linear SVM baseline; none beat it. A perfect selector over the ten candidates would score 0.900, yet every reachable vote lands on 0.771, because model disagreement is *asymmetric*.
3) Calibration plus a rejection option converts a 0.78-accuracy classifier into one that files 81% of articles at 83% accuracy and routes the rest to a human (Section VI). This is the practical contribution: an honest confidence number is more useful than a marginally higher one.

II. Dataset

A Go service polls 97 RSS and Atom feeds into MongoDB and scrapes article bodies. Frozen at `2026-08-26T12:00Z`, the corpus holds 14,189 articles from 40 publishers, 92.6% with a body. Labels are 8,001 single-label human judgements over a flat 13-class taxonomy derived from IPTC Media Topics [1]: *politics*, *business_economy*, *crime_justice*, *technology*, *sport*, *entertainment_arts*, *health*, *education*, *science_space*, *environment_climate*, *disaster_accident*, *conflict_war* and *society_lifestyle*. Class imbalance is $6.7 : 1$. Body coverage on labelled articles is 94.3%, median 3,765 characters, and is uniform across classes (89.7%–99.3%), so no class is disadvantaged by the hypothesis.

A. Preparation

Title, summary and body are cleaned independently. Per-source line-frequency discovery found 355 repeated furniture lines across 47 sources — *"Story continues below this ad"* appears in 75% of one publisher's bodies — including multi-sentence author biographies, which a length-based rule cannot reach. Nine admission gates reject 4.1% of the corpus (implausible timestamps, service bulletins, non-article formats); every gate is a switch so its cost can be measured rather than assumed.

Near-duplicate stories are grouped in two stages, sparse TF-IDF cosine blocking followed by threshold verification with time-gap and boilerplate guards. Calibrated against 43 hand-judged pairs it reaches precision 0.86 at recall 0.81. Grouping on bodies folds 999 articles against 708 for headlines, $+41\%$, which matters because a syndicated copy of a training article appearing in validation is leakage.

III. Method

The final model is a linear SVM over the union of word 1–2 grams (sublinear TF, min_df $= 2$, max_df $= 0.6$) and character `char_wb` 3–5 grams restricted to the first 600 characters, with `class_weight="balanced"` [2], [3]. Input is the title concatenated with the first 4,000 characters of the body; an article with no body falls back to its summary. A sweep

TABLE I

VALIDATION MACRO-F1: THE BODY A/B TEST. EVERY LINEAR MODEL GAINS; COMPLEMENTNB, WHOSE INDEPENDENCE ASSUMPTION IS LENGTH-SENSITIVE, LOSES.

Model	title+summary	title+body	McNemar p
majority baseline	0.025	0.025	—
ComplementNB	**0.635**	0.594	2.8×10^{-2}
TF-IDF + LogReg	0.698	0.752	3.0×10^{-5}
TF-IDF + LinearSVC	0.696	0.753	1.3×10^{-4}
word+char SVM *(final)*	0.712	**0.771**	7.5×10^{-6}

TABLE II

ALTERNATIVES MEASURED AGAINST THE FINAL MODEL ON VALIDATION. NOTHING WAS REJECTED ON INTUITION; EACH ROW CARRIES AN INTERVAL OR A PAIRED TEST.

Alternative	Result
Entity / geography scrubbing (spaCy)	no rule cleared the bar
Up-weighting the title	monotonically worse
C swept over a $6\times$ range	moves macro-F1 by 0.002
XGBoost on 256 SVD components [9]	-0.036, $5\times$ slower
Random forest / extra trees	-0.041 to -0.087
MiniLM embeddings + LogReg [10]	-0.061
Majority vote over 10 models	$+0.001$, $p = 1.00$
Per-class confidence cuts	no better than one global cut

over $\{256, \ldots, \text{full}\}$ characters showed the curve flattening past 2,048; 4,000 was chosen because it has the best publisher-holdout score, not the best validation score. Repeating the title to up-weight it made the model monotonically worse.

A `LinearSVC` decision value is a signed distance from a hyperplane: it has no scale and is not comparable across classes, so it must never be shown as a confidence. Scores are mapped to probabilities by isotonic regression [4] fitted across five grouped folds of the training split, each fold calibrating on data its own base estimator never saw.

IV. EVALUATION METHODOLOGY

Three design choices exist specifically to stop the evaluation flattering itself. First, the train/validation/test split is *grouped and temporal*: cuts are placed on collection time, and no near-duplicate story group spans two splits. Second, 95% intervals are bootstrapped [8] over *story groups* rather than articles, because syndicated copies of one story are not independent observations; model comparisons use McNemar's test [7], not a subtraction of two accuracies. Third, two publishers (*The Hindu*, in-distribution; *The Guardian*, out-of-distribution) are held out and scored on every run, each with the model refitted without them.

The test split was opened exactly once, through a single function that refuses to run without an explicit flag, after the model was frozen — and nothing was changed in response to what it reported.

V. RESULTS

Table I reports the central experiment. The body is worth $+0.059$ macro-F1 on the final model, with disjoint bootstrap

TABLE III

THE HELD-OUT TEST SPLIT, OPENED ONCE. VALIDATION SHOWN FOR COMPARISON.

	validation	test
articles	1,120	1,159
macro-F1	0.771 [0.743, 0.796]	**0.751 [0.719, 0.780]**
accuracy	0.794	0.778
expected calibration error	0.039	**0.021**
shipping policy, cut $= 0.584$		
coverage (articles filed)	0.761	0.807
accuracy on filed articles	0.879	**0.834**
accuracy if forced to answer	0.795	0.777

intervals. Against a rung reproducing an earlier headline-only system on the same snapshot (0.696) the gain is $+0.075$. The one model that gets *worse* with more text is ComplementNB, whose term-independence assumption is sensitive to document length — a useful reminder that "more text" is not universally better.

Publisher generalisation *improved* rather than degraded, which was the main risk the hypothesis carried: *The Hindu* rose $0.682 \rightarrow 0.769$ and *The Guardian* $0.665 \rightarrow 0.724$ when bodies were added. The boilerplate work in preparation is the likely reason.

Table II lists what did not work. The ensemble row deserves comment: an oracle that could always pick the correct member among the ten candidates would score 0.900 $[0.878, 0.919]$, disjoint from 0.771, so the models genuinely disagree. Every vote still lands within 0.002 of the incumbent because the disagreement is asymmetric — the embedding model rescues 70 of the incumbent's errors while destroying 148 of its correct answers, and a vote has no way to know which side of that trade it is on.

VI. CALIBRATION AND ABSTENTION

Calibration costs almost nothing in accuracy (0.769 against a raw 0.771) and buys a probability that can be acted on. Fig. 1(a) shows the model is under-confident rather than over-confident, the unusual direction; expected calibration error [5] is 0.039 on validation and 0.021 on test, so the confidence number generalised *better* than the accuracy did.

That makes classification with rejection [6] straightforward. One global threshold suffices — per-class thresholds, which exist because raw scores are not comparable across classes, bought nothing once calibration had made them comparable (0.879 against 0.891 at matched coverage). Table III gives the outcome on the held-out test split: declining the least certain 19.3% of articles raises accuracy on the remainder from 77.7% to 83.4%.

The gap between validation and test is -0.021 against a pre-registered guard of ± 0.05, with overlapping intervals. Only *conflict_war* moved further ($0.74 \rightarrow 0.60$); it is the thinnest class in the corpus at 120 training articles, and its validation interval $[0.62, 0.83]$ had already flagged it as unreliable.

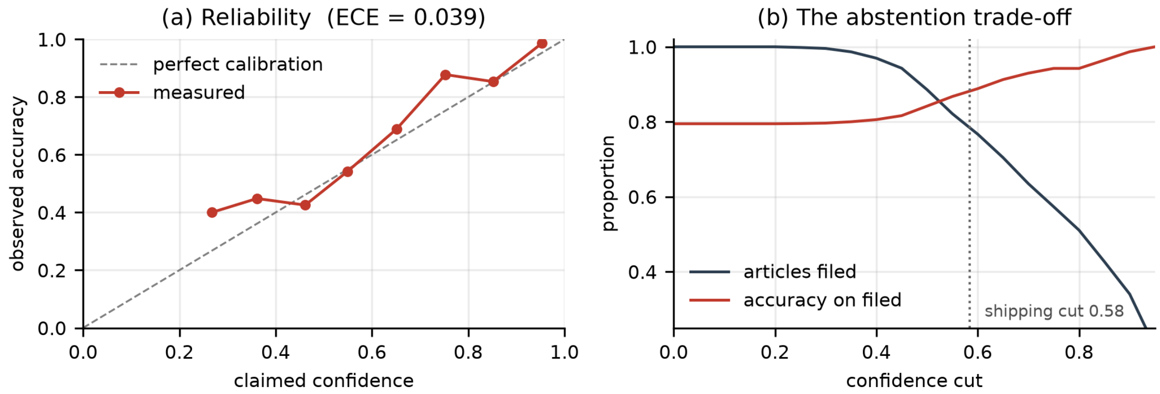

Fig. 1. (a) Reliability on validation. Points above the diagonal mean the model is *under*-confident — in the 0.7–0.8 band it claims 0.75 and delivers 0.88 — which is the safe direction for a system that abstains. (b) The rejection trade-off: raising the cut files fewer articles but is right more often on those it files. The shipping cut of 0.584 was fitted on training out-of-fold probabilities for 90% precision and never on anything it is scored against.

VII. Discussion and Limitations

Roughly 18.6% of validation errors fall on the three class pairs where human annotators themselves disagreed when the corpus was labelled, an estimate that matches an independent measurement of label disagreement across multi-article story groups (18–22%). Macro-F1 therefore has a real ceiling well below 1.0, and chasing the remaining confusions would mean fitting the annotation noise rather than the task.

society_lifestyle scores F1 0.42: it is a definitional grab-bag of community, labour and lifestyle reporting. Notably it still calibrates honestly, reaching 91% precision at a high cut — the class is weak at *ranking* but not at *knowing*, so abstention protects it where an earlier system had retired it entirely. The collection window spans four days, so nothing here measures topic drift over weeks; the publisher holdouts carry the whole generalisation argument. The system is English-only and India-heavy. Serving costs 0.78 ms per article.

VIII. Conclusion

On a hand-labelled 13-class news corpus, article bodies beat headlines by +0.059 macro-F1, and no fancier model family or ensemble improved on a linear SVM over TF-IDF — the gains came from data preparation and honest evaluation, not from model capacity. Adding calibrated confidence and a rejection option turned a 77.7%-accurate classifier into one that files 80.7% of unseen articles at 83.4% accuracy, which is the form in which such a system is actually usable. Future work is a fine-tuned transformer encoder, deliberately sequenced after this A/B so that any gain can be attributed.

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